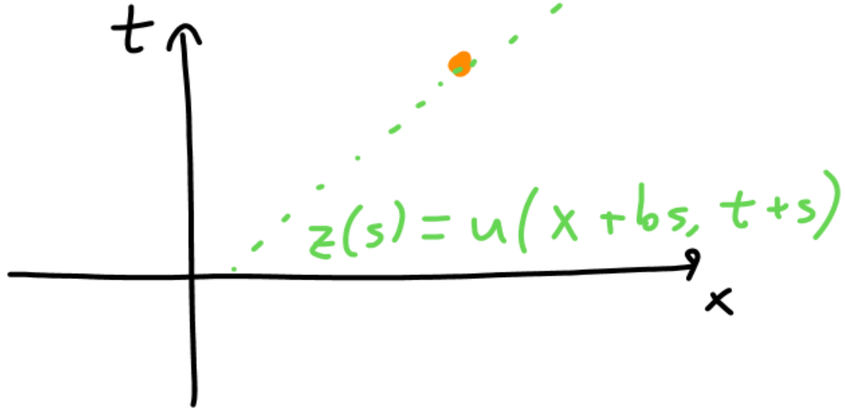
HW 1 : posted tomorrow, due Sept 3.

Recall:

$$\begin{cases} u_t + b \cdot Du = 0, & u(x,t), \quad t \geq 0, \quad x \in \mathbb{R}^n. \\ u(x,0) = g(x) \end{cases}$$



Last time: 1D case.

General case:

$$z(s) = u(x + bs, t + s) = u(x_1 + b_1 s, x_2 + b_2 s, \dots, x_n + b_n s, t + s)$$

$$\frac{d}{ds} z(s) = u_{x_1} b_1 + \cdots + u_{x_n} b_n + u_t = u_t + b \cdot Du = 0$$

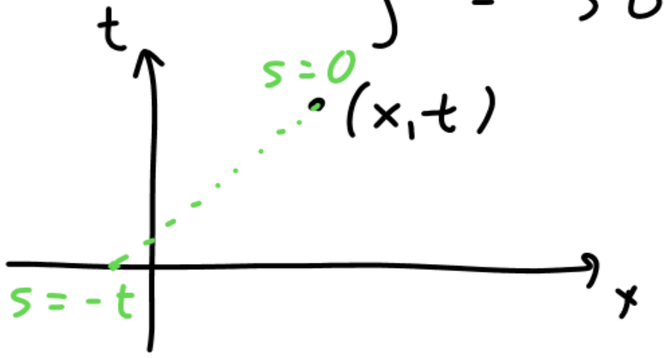
$\Rightarrow z$ is constant in s. $\quad z(0) = z(-t) \rightarrow \boxed{u(x,t)} = u(x - tb, 0)$

$$= g(x - tb)$$

Non-homogeneous linear transport:

$$\begin{cases} u_t + b \cdot Du = f(x,t) \\ u(x,0) = g(x) \end{cases}$$

f = source



Try same approach:

$$z(s) = u(x+bs, t+s)$$

$$\frac{d}{ds} z(s) = u_t(x+bs, t+s) + b \cdot Du(x+bs, t+s) = f(x+bs, t+s)$$

(z no longer constant in s.)

Integrate $\frac{d}{ds} z(s)$ in s.

Fund. Thm of Calc: $\int_{-t}^{0} \frac{d}{ds} z(s)\, ds = z(0) - z(-t) = u(x,t) - g(x-tb$

$$u(x,t) - g(x-tb) = \int_{-t}^{0} \frac{d}{ds} z(s)\, ds$$

$$= \int_{-t}^{0} f(x+sb, t+s)\, ds$$

$$\rightarrow \boxed{u(x,t) = g(x-tb) + \int_{-t}^{0} f(x+sb, t+s)\, ds}$$

$v = -s$

$$u(x,t) = g(x-tb) + \int_{0}^{t} f(x-vb, t-v)\, dv$$

HW: Same PDE on domains with boundaries.

Later: b replaced with $a(u)$

Laplace's Equation

$$\Delta u = 0 \qquad \text{(Laplace's Equation)}$$

$$\Delta u = f \qquad \text{(Poisson's Equation)}$$

Recall: $\Delta u = \sum_{i=1}^{n} u_{x_i x_i} = \operatorname{tr}(D^2 u)$

u solving $\Delta u = 0$ is called <u>harmonic</u>.

These PDEs are <u>stationary</u> (no time dependence)

Physical interpretation:

$u(x)$ is a density

(or temperature, chemical concentration, electric potential, etc....)

If u is in equilibrium, flux density $= 0$

For a 3D region $V \subset \mathbb{R}^3$,

let $\vec{j}$ = flux density,

let ∂V = boundary of V

$$0 = \int_{\partial V} \vec{j} \cdot \nu \, dS$$

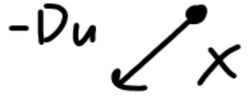
normal vector

$$= \int_V \operatorname{div} \vec{j} \, dx \quad \text{(Div. Thm)}$$

Since V is any region, one can show

$$\operatorname{div}(\vec{j}) = 0.$$

If $\vec{j} = -aDu \qquad a > 0$



(particles diffusing from regions of high concentration to regions of low concentration.)

$$\operatorname{div}(-aDu) = 0 \rightarrow \operatorname{div}(Du) = 0$$

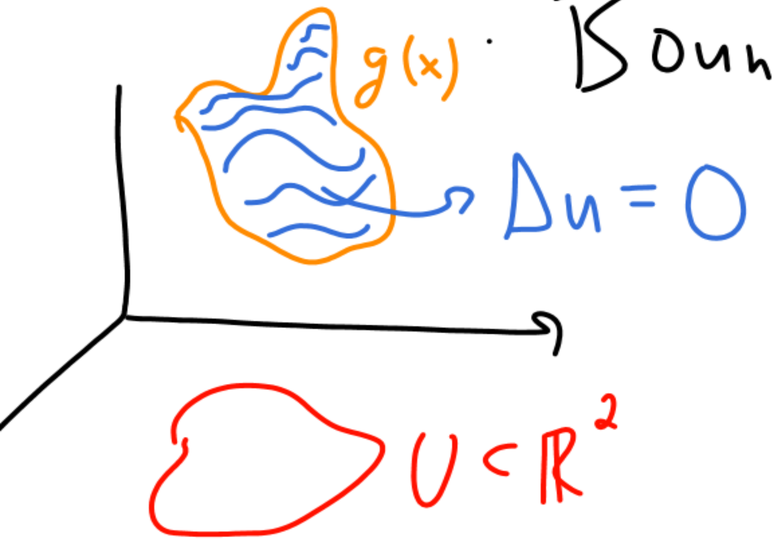
Claim: $\operatorname{div}(Du) = \Delta u$.

$$\operatorname{div}(Du) = \sum_{i=1}^{n} \frac{\partial}{\partial x_i}\left((Du)_i\right) \qquad Du = (u_{x_1}, \dots, u_{x_n})$$

$$= \sum_{i=1}^{n} \frac{\partial}{\partial x_i}\left(u_{x_i}\right) = \sum_{i=1}^{n} u_{x_i x_i} = \Delta u \quad \checkmark$$

We'll study these equations on:

- All of $\mathbb{R}^n$
- Bounded subsets $U \subset \mathbb{R}^n$.



$\Delta u = 0$ shows up in complex analysis (HW)

Fundamental Solution:

Want to write solutions to $-\Delta u = f$

as $$\boxed{u(x) = \int_{\mathbb{R}^n} \Phi(x,y) f(y)\, dy}$$

Formally, $\Phi(x,y)$ solves $-\Delta_x \Phi(x,y) = \delta_y(x)$

↳ Dirac delta

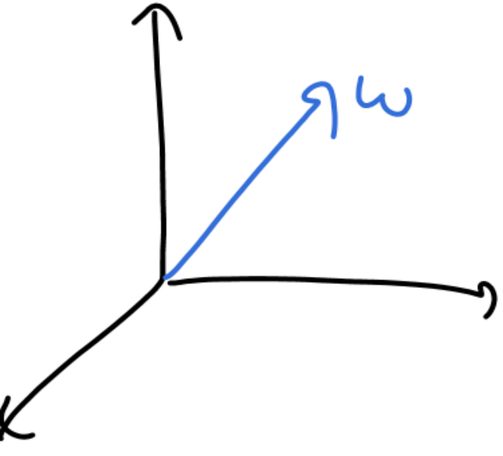
If $-\Delta \Phi(x,0) = \delta_0(x)$,

then $-\Delta \Phi(x,0) = 0$ outside $x=0$

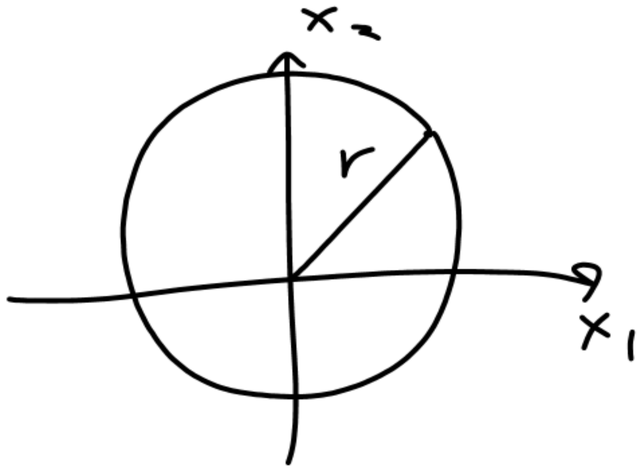
Fact: The operator $-\Delta$ is "rotationally invariant," i.e. $-\Delta u(Rx) = -\Delta u(x)$

for any rotation matrix R

$n=2$ case: $R = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$



Let's look for a fundamental solution that's rotationally invariant:



Let $r = \sqrt{x_1^2 + \cdots + x_n^2}$

Look for $\Phi(x,0) = v(r)$

Want $\Delta_x \Phi(x,0) = 0 \qquad x \neq 0$

Compute this in terms of v

$$\Phi_{x_i} = \frac{\partial}{\partial x_i} v(r) = \frac{\partial v}{\partial r}\frac{\partial r}{\partial x_i} \qquad \text{(Chain Rule)}$$

$$r^2 = x_1^2 + \cdots + x_n^2 \longrightarrow 2r\frac{\partial r}{\partial x_i} = 2x_i \longrightarrow \frac{\partial r}{\partial x_i} = \frac{x_i}{r}$$

$$\boxed{\Phi_{x_i} = \frac{x_i}{r}\frac{\partial}{\partial}}$$

$$\Phi_{x_i} = \frac{\partial v}{\partial r}\frac{x_i}{r}$$

$$\Phi_{x_i x_i} = \frac{\partial^2 v}{\partial r^2}\frac{\partial r}{\partial x_i}\frac{x_i}{r} + \frac{\partial v}{\partial r}\left(\frac{r - x_i \frac{\partial r}{\partial x_i}}{r^2}\right)$$

$$= \frac{\partial^2 v}{\partial r^2}\frac{x_i^2}{r^2} + \frac{\partial v}{\partial r}\left(\frac{1}{r} - \frac{x_i^2}{r^3}\right)$$

$$\Delta \Phi = \sum_{i=1}^{n} \Phi_{x_i x_i} = \frac{\partial^2 v}{\partial r^2}\frac{1}{r^2}\sum_{i=1}^{n} x_i^2 + \frac{\partial v}{\partial r}\left(\frac{n}{r} - \frac{1}{r^3}\sum_{i=1}^{n} x_i^2\right)$$

$$= \frac{\partial^2 v}{\partial r} + \frac{\partial v}{\partial r}\left(\frac{n}{r} - \frac{1}{r}\right) = v''(r) + \frac{n-1}{r}v'(r)$$

$v(r)$ solves $v''(r) + \frac{n-1}{r} v'(r) = 0$.

let $w = v'$:

$$w' + \frac{n-1}{r} w = 0$$

$$w' = -\left(\frac{n-1}{r}\right) w$$

$$\frac{w'}{w} = -\frac{(n-1)}{r}$$

$$\ln|w| = c_1 - (n-1)\ln(r)$$

$$w = e^{c_1} e^{-(n-1)\ln(r)}$$

$$= c_0 r^{-(n-1)}, \text{ for some } c_0.$$

$$w = \frac{c_0}{r^{n-1}} = v'(r)$$

Integrate in r: If $n \geq 3$,

$$v(r) = a - \frac{c_0}{r^{n-2}} \cdot \frac{1}{n-2}$$

$$= a - \frac{b}{r^{n-2}} \qquad b = \frac{c_0}{n-2}$$

Choose $a = 0$,

$$v(r) = \frac{b}{r^{n-2}}$$

$$\Phi(x, 0) = \frac{1}{|x|^{n-2}} \frac{1}{n(n-2)\alpha(n)}$$

$\alpha(n)$ = surface area of n-sphere

In general, $\Phi(x,y) = \dfrac{C_n}{|x-y|^{n-2}}$ $\quad |x-y| = \sqrt{(x_1-y_1)^2 + \cdots + (x_n-y}$

$v'(r) = \dfrac{c_0}{r^{n-1}}$ $\quad n=2$: $\quad v'(r) = \dfrac{c_0}{r}$

$$v(r) = a + c_0 \ln(r)$$

Choose $a=0$, $\quad \Phi(x,y) = \dfrac{-1}{2\pi} \ln|x-y|$

How to use this fund. solution:

<u>Thm</u> If $f \in C_c^2(\mathbb{R}^n)$, then

$u(x) = \int_{\mathbb{R}^n} \Phi(x-y) f(y)\,dy$ is $C^2(\mathbb{R}^n)$ and solves $-\Delta u = f$.

Notation: a function $g \in C^2(\mathbb{R}^n)$ if all 1st & 2nd-order partial derivatives exist & are continuous in $\mathbb{R}^n$.

$g \in C_c^2(\mathbb{R}^n)$ if $g \in C^2$ and $g = 0$ outside some bounded set $U \subset \mathbb{R}^n$.